

Interpreting Deepfake Detection on SID-Set

Foundation-Model Baselines and Mechanistic Analysis
via Sparse Autoencoders

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April 2026

Abstract

We present a two-part interpretability study of social-media deepfake detection on the SID-Set benchmark, which frames the problem as three-class classification of *real*, *fully synthetic*, and *tampered* images. **Part I** establishes transparent baselines: we freeze three vision foundation models (CLIP ViT-B/32, DINOv2-Base, SigLIP-Base) and train simple linear probes on top, achieving 90.0% macro-F1 for classification and 0.375 mean IoU for tampered-region localization with no fine-tuning and no GPU at evaluation time. A suite of eight interpretability analyses, L1/L2 sparsity, decision trees, SHAP attribution, Token-CAM, CKA similarity, calibration, layer/pooling ablations, and robustness testing, shows that deepfake cues are class-specific, concentrated in the last quarter of the backbone, and remarkably stable under social-media degradations. **Part II** opens the black box of a state-of-the-art detector: we apply sparse autoencoders (SAEs) to the 4096-dimensional classification embedding of SIDA-7B, a multimodal deepfake detector. We compare five SAE variants (Vanilla, TopK, Gated, JumpReLU, Matryoshka), demonstrate causal feature steering that shifts per-class accuracy by up to 9.4 percentage points, identify a sharp decision-boundary cliff where the hardest 20% of tampered images collapse to 12.7% accuracy, and map per-class adversarial vulnerability. Strikingly, both parts converge on the same conclusions: tampered images are the hard class, synthetic-discriminative features are concentrated and high-amplitude, and deepfake detection relies on globally distributed rather than spatially localized cues. Together the two analyses provide a complete interpretability picture, from raw pretrained features to fine-tuned model internals, of what makes AI-generated images detectable and where current detectors are most fragile.

1 Introduction

Modern generative systems such as FLUX [28] and latent diffusion inpainting pipelines [29] can produce images indistinguishable from real photographs, even to trained experts [1]. SID-Set [1] poses detection as a *three-class* problem: *real* photographs vs. *fully synthetic* images (entirely AI-generated) vs. *tampered* images (real images with AI-inpainted regions). This framing reflects real-world deployment where the forgery type is unknown *a priori*, and it exposes a fundamental asymmetry between the two forgery classes: synthetic images carry a pervasive generator signature, while tampered images are predominantly authentic with only a localized disruption.

Recent work [7, 8, 10] has shown that vision foundation models (VFMs), particularly CLIP, serve as effective feature extractors for deepfake detection. However, this literature has notable gaps. First, nearly all work operates in a binary (real vs. fake) setting and cannot separate the two forgery regimes. Second, most studies examine a single backbone. Third, existing methods treat foundation models as black boxes: there is no systematic investigation of *what* frozen representations encode about deepfakes, *where* in the network that information resides, or *how* the decision is formed. Fourth, many papers claiming to use “frozen” features still add

adapters, prompt tuning, or LoRA, preventing clean analysis of the pretrained representations alone. Finally, for the state-of-the-art models that *do* fine-tune, we have no mechanistic picture of the internal computation.

This report addresses these gaps through two complementary studies on the same SID-Set benchmark:

- **Part I (Frozen Foundation-Model Baselines).** We establish transparent linear baselines on three backbones spanning the major pretraining paradigms and conduct a comprehensive interpretability suite. This answers the question: *what is already in the pretrained features, and through what mechanism* does a simple classifier exploit it?
- **Part II (Mechanistic Analysis of SIDA via Sparse Autoencoders).** We take SIDA-7B [1], a 7-billion-parameter multimodal deepfake detector, and decompose its 4096-dimensional classification embedding using five SAE variants. This answers the question: *how does the fine-tuned state-of-the-art model actually reach its decision, and where is it most fragile?*

Contributions. Our contributions are:

- A systematic, interpretable comparison of three frozen backbones (CLIP ViT-B/32, DINOv2-Base, SigLIP-Base) on the three-class SID-Set formulation, covering both classification and patch-level localization, with no fine-tuning.
- A comprehensive interpretability suite for the frozen-feature regime: L1/L2 sparsity, decision trees, SHAP, Token-CAM, CKA, calibration, pooling and layer ablations, and robustness testing.
- The first SAE-based mechanistic analysis of a multimodal deepfake detector, comparing five SAE variants and demonstrating causal feature steering.
- Identification of a critical failure mode: hard tampered samples (bottom 20% confidence) achieve only 12.7% accuracy with mean confidence 0.21.
- A per-class adversarial vulnerability map showing that the *most accurate* class (fully synthetic, 99.9%) is also the *most brittle* to targeted feature perturbations.
- A bridging analysis connecting the two parts: we show that the frozen-feature and SAE analyses converge on the same qualitative conclusions, validating both.

2 Related Work

2.1 Deepfake Detection Datasets and Traditional Methods

Early datasets, ForgeryNet [18] and DFFD [19], focused on facial manipulation, while later benchmarks such as GenImage [20] and AIGCD [21] expanded to general images but relied on older generators, making detection artificially easy. Most prior benchmarks frame detection as binary and do not distinguish synthetic from tampered content. SID-Set [1] addresses these limitations with 300K images generated using FLUX and latent diffusion inpainting, three-class labels, and pixel-level masks.

Traditional detectors exploit generator-specific artifacts: CnnSpot [22] learns CNN texture patterns, FreDect [23] and Fusing [24] operate in the frequency domain, Gram-Net [25] captures texture statistics, and LGrad [26]/LNP [27] use gradient and noise-pattern representations. These methods achieve strong in-domain performance but generalize poorly across generators, achieving only 37–55% accuracy on SID-Set with pretrained weights [1].

2.2 Foundation Models for Deepfake Detection

Ojha et al. [7] (UnivFD) demonstrated that frozen CLIP features with a linear classifier can detect AI-generated images and generalize across unseen generators. This catalyzed a line of work

adapting CLIP for detection: prompt tuning [8], category-specific prompt injection [9], trainable feature guidance modules [10], and LayerNorm tuning [11]. For localization, DeCLIP [12] decodes frozen CLIP patch embeddings with a learned convolutional decoder, though it showed that CLIP struggles with locally manipulated images. RIGID [13] leverages DINOv2’s perturbation sensitivity for training-free detection, and Cocchi et al. [14] studied linear probes on CLIP/DINO features under image transformations.

For explanation rather than mechanism, M2F2-Det [15], ForgeryGPT [16], and FakeShield [17] use LLMs to generate textual rationales for detection decisions. These provide post-hoc natural-language outputs whose faithfulness to the model’s internal decision process is not guaranteed. No prior work has performed mechanistic analysis of *which* embedding dimensions, *at what depth*, and *via what spatial mechanism* frozen representations encode deepfake cues, nor have SAEs been applied to a multimodal deepfake detector. Our two-part study fills both gaps.

2.3 Mechanistic Interpretability and Sparse Autoencoders

Mechanistic interpretability [39, 31, 32] aims to reverse-engineer neural network internals into human-understandable components. Sparse autoencoders decompose a dense activation vector $\mathbf{x} \in \mathbb{R}^d$ into a sparse code $\mathbf{z} \in \mathbb{R}^m$ (with $m \gg d$) such that $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{z} + \mathbf{b}$ approximates $\mathbf{x}$. The overcomplete basis lets each feature represent a more monosemantic concept than raw dimensions [32]. Sparsity is enforced via L1 penalties [31], top- k selection [33], gating [34], learnable thresholds (JumpReLU) [35], or nested structure (Matryoshka) [36]. SAEs have been extensively applied to language models; their application to multimodal detectors, and to deepfake detection specifically, remains open prior to this work.

3 Dataset

Both parts of this study use a balanced subset of SID-Set [1]. Tampered images carry pixel-level binary masks, enabling the localization experiments in Part I. Tampered images were generated by inpainting objects into real images using latent diffusion models; the fully synthetic class spans multiple generators (primarily FLUX); real images are sourced from COCO, OpenImages, and Flickr30k.

Table 1: Dataset splits used in this study. Part I uses a 10k balanced subset; Part II uses the 60k test partition released with SIDA-7B embeddings (20k per class).

Split	Real	Synthetic	Tampered
<i>Part I (frozen-backbone experiments)</i>			
Train	2,667	2,667	2,666
Val	333	333	334
Test	333	334	333
<i>Part II (SAE experiments on SIDA embeddings)</i>			
SAE Train	10,000	10,000	10,000
SAE Eval	10,000	10,000	10,000

PART I

Interpretable Frozen Foundation-Model Baselines

In Part I we ask what information about deepfakes is already present in the representations of modern vision foundation models, before any fine-tuning. Rather than training end-to-end detectors, we adopt an *embedding-first, interpretable* approach: we freeze backbone weights, extract a single fixed feature vector per image, and train transparent classifiers on top. This design isolates what the pretrained features already encode, enables interpretability through coefficient analysis and spatial attribution, and establishes reproducible baselines that require no GPU for evaluation after a one-time feature extraction step.

4 Approach

4.1 Frozen Backbone Feature Extraction

We extract fixed representations from three complementary vision backbones, all based on the ViT-Base architecture but spanning the major pretraining paradigms.

Table 2: Backbone configurations. All weights are frozen; no fine-tuning is performed.

Backbone	Training	Patch	Grid	Global D
CLIP ViT-B/32	Contrastive (text-image)	32	7×7	512 (proj.)
DINOv2-Base	Self-supervised (DINO)	14	16×16	768 (CLS)
SigLIP-Base	Sigmoid contrastive	16	14×14	768 (mean-pool)

CLIP uses language-supervised contrastive learning, DINOv2 uses self-supervised distillation with no text signal, and SigLIP uses a sigmoid-based vision-language objective. The diversity in pretraining allows us to test which class of pretext task produces representations most useful for forensic tasks.

4.2 Classification Pipeline

For each backbone we extract the default global embedding (projection head for CLIP, CLS token for DINOv2, mean-pooled patches for SigLIP) for all 10k images and train three classifiers: (i) **L2 logistic regression**, a standard multinomial logistic regression with ℓ_2 penalty; (ii) **L1 logistic regression**, the ℓ_1 -penalized variant that performs automatic feature selection; and (iii) **decision trees**, depth-tuned on validation (depths 3–10) as human-readable baselines.

4.3 Localization Pipeline

For tampered images we extract *patch tokens*, the spatial token grids, and train a per-patch SGD classifier to distinguish tampered from non-tampered patches. Predicted patch scores are bilinearly upsampled to 224×224 and compared against ground-truth binary masks using IoU, Dice, and AUC.

5 Classification Results

Table 3: Classification results on the test set. All classifiers use frozen global embeddings.

Backbone	Classifier	Test F1	Test Acc
CLIP ViT-B/32	LogReg-L2	0.8898	0.890
	LogReg-L1 (277/512 dims)	0.8999	0.900
	DecTree (depth=10)	0.6907	0.691
DINOv2-Base	LogReg-L2	0.8612	0.862
	LogReg-L1 (505/768 dims)	0.8666	0.867
	DecTree (depth=10)	0.5861	0.585
SigLIP-Base	LogReg-L2	0.8566	0.857
	LogReg-L1 (324/768 dims)	0.8558	0.856
	DecTree (depth=10)	0.6625	0.662

All three backbones achieve 86–90% macro-F1 with simple linear classifiers on frozen features. CLIP leads despite having the smallest embedding dimension (512-d projected), likely because its contrastive training with language produces a more compressed, discriminative representation for this task. L1 regularization matches or slightly outperforms L2 while using substantially fewer dimensions: CLIP retains only 277/512 dimensions (54%), DINOv2 505/768 (66%), and SigLIP 324/768 (42%), revealing significant redundancy in the embedding space for deepfake detection.

Decision trees perform dramatically worse (59–69% F1). This is itself an informative negative result: the class boundary cannot be captured by axis-aligned if-then rules. The decision surface is genuinely linear but not axis-aligned; a soft weighted combination of many dimensions is essential. For context, SIDA [1], the 7B-parameter model fine-tuned on the full 300K-image dataset, achieves 93.5% F1. Our best frozen-feature classifier reaches roughly 96% of this performance with a model several thousand times smaller and fully transparent.

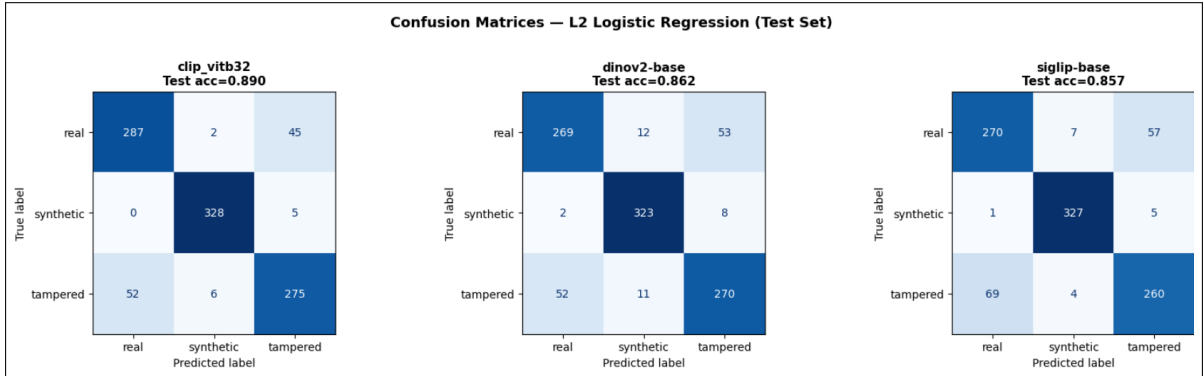


Figure 1: Confusion matrices (L2 LogReg, test set). The tampered–real boundary shows the most confusion across all three backbones, foreshadowing the “hard tampered” failure mode we will analyze mechanistically in Part II.

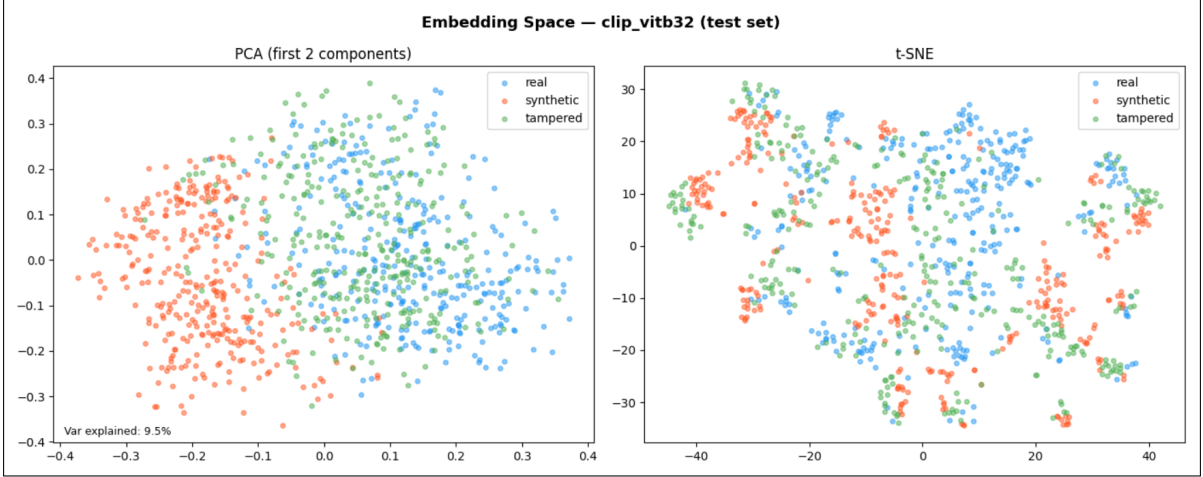


Figure 2: PCA (left) and t-SNE (right) of CLIP test embeddings. Synthetic images form a distinct cluster; real and tampered overlap substantially, consistent with the confusion-matrix evidence.

6 Classification Interpretability

6.1 L1 Coefficient Sparsity and Feature Selection

The L1 classifier reveals which embedding dimensions drive each class prediction. The *synthetic* class exhibits dramatically larger coefficients than real or tampered: for DINOv2, the top synthetic coefficient reaches +41.13 (dimension 5), while the largest real coefficient is only ± 9.8 (CLIP). This indicates that fully synthetic images produce a concentrated, high-amplitude signal that is easy for a linear probe to detect. The *tampered* class uses more distributed, moderate-weight features, consistent with the fact that tampered images are mostly real except for localized inpainted regions. The *real* class coefficients are largely negative, defined by the *absence* of synthetic/tampered indicators. This pattern (concentrated synthetic features, diffuse tampered features) will reappear almost exactly in the SAE selectivity analysis of Part II.

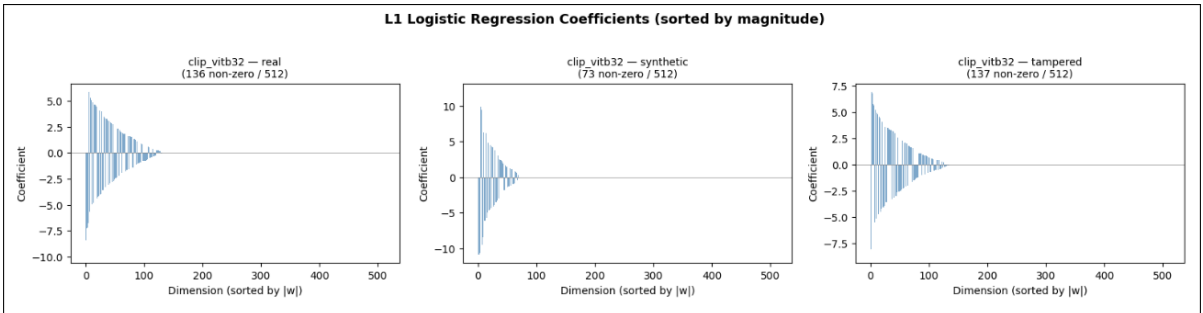


Figure 3: L1 coefficient sparsity. Synthetic-class coefficients are sparse and high-magnitude; real and tampered use more distributed, lower-weight features.

6.2 Decision Tree Analysis

Despite poor test accuracy, decision trees provide interpretive value. The tree’s top-split dimensions (d428 for CLIP, d5/d540 for DINOv2) match the top L1 coefficients, cross-validating both methods. However, the severe val-to-test gap (e.g., DINOv2: val F1=0.88, test F1=0.59) confirms that tree-like decision rules catastrophically overfit on high-dimensional embedding spaces.

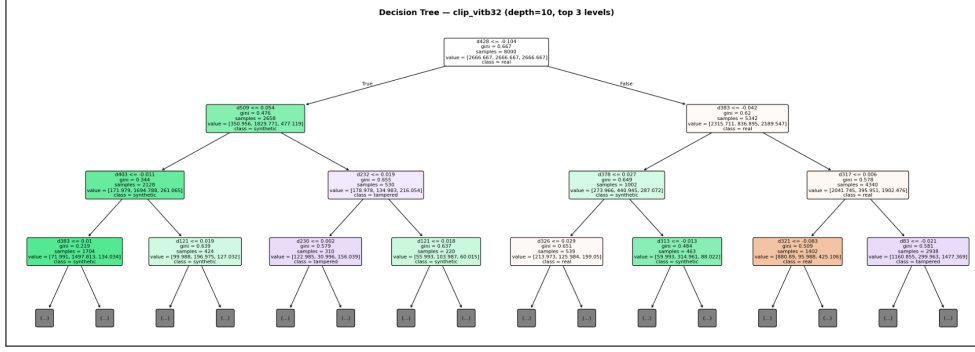


Figure 4: Decision tree top-3 levels. The root split on d428 (CLIP) matches the top L1 coefficient, but deeper splits overfit.

6.3 SHAP Attribution

For a linear classifier, exact SHAP values are $\phi_j = w_j \cdot (x_j - \mathbb{E}[x_j])$, providing per-image, per-dimension attribution without approximation. A shortcut analysis reveals that the top-20 most important dimensions are largely *class-specific*: across all three backbones, only 2–3 dimensions appear in the top-20 for all three classes simultaneously, while 8–14 dimensions are unique to a single class. This provides quantitative evidence against shortcut learning: the classifier uses genuine per-class cues rather than relying on shared spurious features.

Table 4: SHAP shortcut analysis. Overlap = dimensions in top-20 for all 3 classes. Low overlap indicates class-specific, non-shortcut features.

Backbone	Shared (all 3)	Real-only	Synth-only	Tamp-only
CLIP ViT-B/32	2 dims	8	4	4
DINOv2-Base	3 dims	8	6	11
SigLIP-Base	3 dims	9	6	14

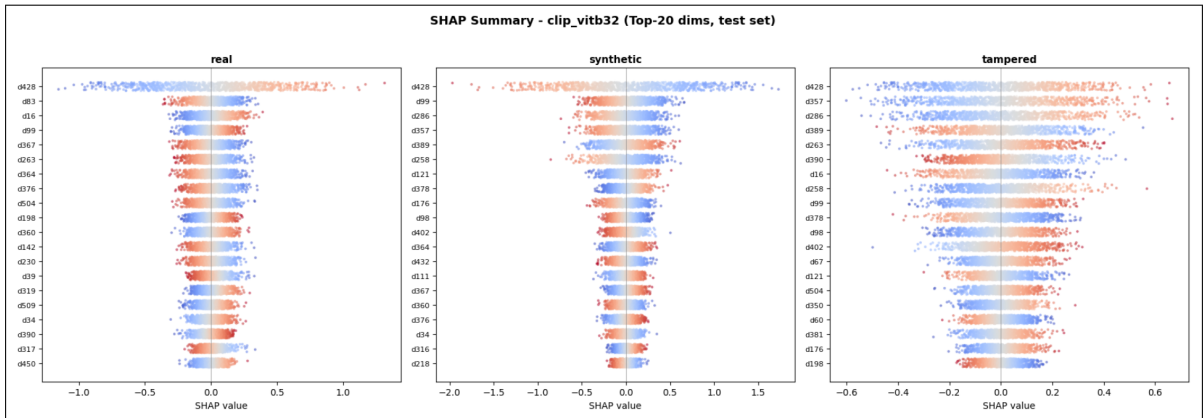


Figure 5: SHAP beeswarm summary. Each dot is a test image; color indicates feature value (blue=low, red=high). Dimension d428 (CLIP) dominates all classes but with opposite signs, explaining its “shared” status.

6.4 Calibration (ECE)

Table 5: Expected Calibration Error (ECE) on the test set. Lower is better.

Backbone	ECE
CLIP ViT-B/32	0.0653
DINOv2-Base	0.0890
SigLIP-Base	0.0627

All backbones achieve ECE below 0.09, indicating reasonably well-calibrated predictions. SigLIP is the best-calibrated (0.063) despite having slightly lower accuracy, suggesting its mean-pooled features produce more uniformly confident predictions.

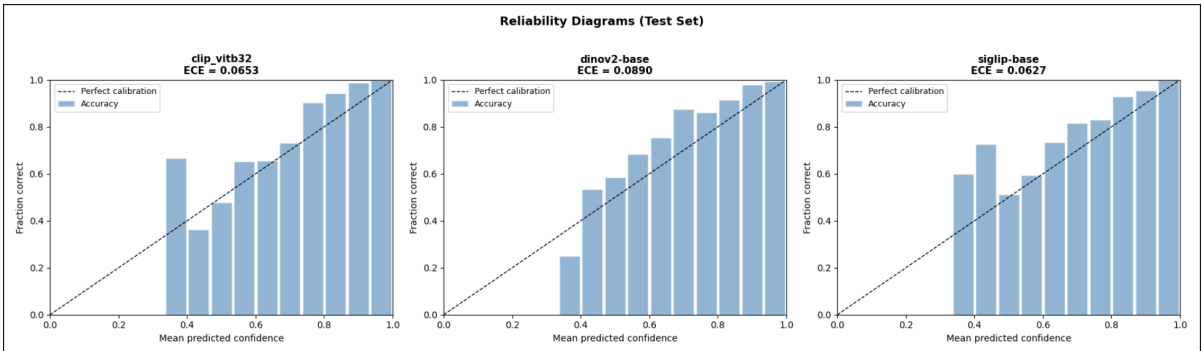


Figure 6: Reliability diagrams. Bars show accuracy per confidence bin; the dashed line is perfect calibration. All backbones show slight overconfidence at high confidence levels.

7 Layer Probing: Where Do Deepfake Cues Emerge?

Table 6: Classification F1 by backbone layer (L2 LogReg on CLS/mean-pool from each layer). L-1 is the final layer; L-12 is twelve layers before the end.

Backbone	L-12	L-8	L-4	L-1	Concat(L-1,-2,-3)
CLIP ViT-B/32	0.521	0.846	0.908	0.902	<u>0.919</u>
DINOv2-Base	0.528	0.816	0.870	0.838	<u>0.902</u>
SigLIP-Base	0.685	0.852	0.857	0.851	<u>0.876</u>

Three findings emerge. First, deepfake cues emerge primarily in the last quarter of the backbone: the jump from L-12 to L-8 is large (+30 F1 points), while L-8 to L-4 adds another 2–6 points. Second, for CLIP and DINOv2, layer L-4 actually *outperforms* the final layer L-1, suggesting that late layers specialize for pretraining objectives (contrastive matching, self-supervised prediction) and slightly dilute the forensic signal. Third, concatenating the last three layers consistently provides a 2–4 point boost over any single layer, confirming that complementary information is distributed across adjacent layers.

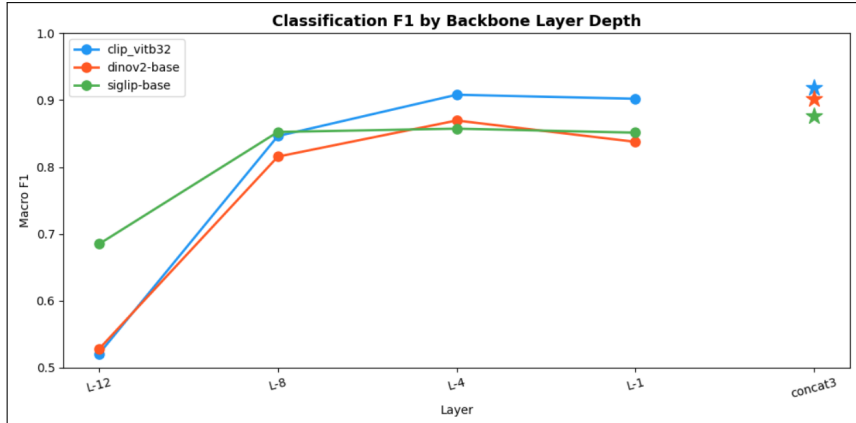


Figure 7: Classification F1 vs. layer depth. Stars denote concat(L-1,-2,-3). For CLIP and DINOv2, intermediate layer L-4 outperforms the final layer.

8 Pooling Ablation

Table 7: Default embedding vs. mean-pooled patch tokens (768-d for all).

Backbone	Default Pooling	Default F1	Mean-Pool F1
CLIP ViT-B/32	Projection (512-d)	0.890	0.807
DINOv2-Base	CLS token (768-d)	0.861	0.871
SigLIP-Base	Mean-pool (768-d)	0.857	0.851

CLIP’s projection head compresses 768-d hidden states to 512-d, and this compression actually *helps*: bypassing it (mean-pooling raw 768-d tokens) drops F1 by 8.3 points. For DINOv2, mean-pooling marginally outperforms the CLS token (+1.0 F1), suggesting that spatial information averaged across patches captures complementary cues missed by the CLS summary. SigLIP shows no meaningful difference, as expected since its default embedding is already mean-pooled.

9 Robustness to Social-Media Transforms

Table 8: Classification F1 under common image degradations (test set).

Backbone	Clean	JPEG-50	JPEG-75	Resize-half	Blur- $\sigma 2$
CLIP ViT-B/32	0.890	0.877	0.869	0.883	0.861
DINOv2-Base	0.861	0.856	0.861	0.852	0.856
SigLIP-Base	0.857	0.860	0.863	0.857	0.862

All backbones show remarkable robustness. The worst degradation is -2.9 F1 points (CLIP under blur), and DINOv2 is nearly invariant to all transforms (< 0.9 point change). SigLIP actually *improves* slightly under JPEG and blur, possibly because these degradations smooth out fine details that cause some real/tampered confusion. This stability is encouraging for deployment on social media, where images routinely undergo recompression and resizing, and it argues that the detection signal resides in high-level semantic features rather than fragile frequency-domain artifacts.

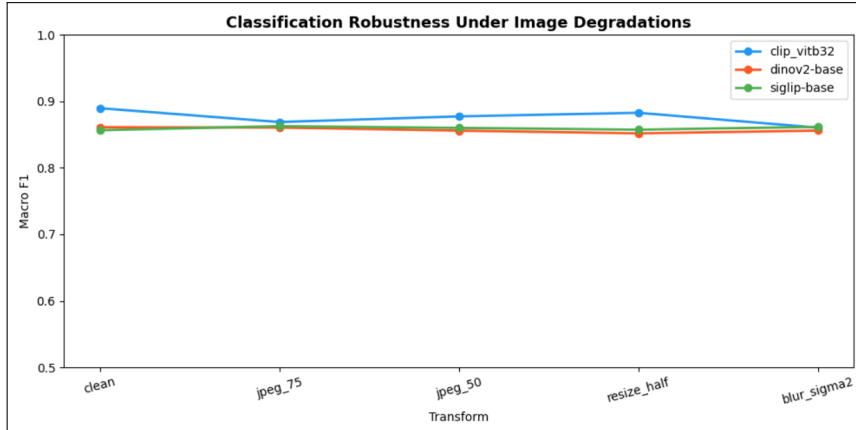


Figure 8: Classification F1 under image degradations. All backbones degrade by $<3\%$ even under aggressive JPEG-50 compression.

10 CKA Representation Similarity

Table 9: Linear CKA between backbone test embeddings.

	CLIP	DINOv2	SigLIP
CLIP	1.000	0.538	0.748
DINOv2	0.538	1.000	0.582
SigLIP	0.748	0.582	1.000

CLIP and SigLIP are the most similar (CKA=0.748), expected since both are trained with vision-language contrastive objectives. DINOv2, the only self-supervised backbone, is the most distinct (CKA ≤ 0.58 with both VLMs). This suggests that an ensemble combining DINOv2 with either VLM backbone could capture complementary representations, consistent with our layer-concatenation result.

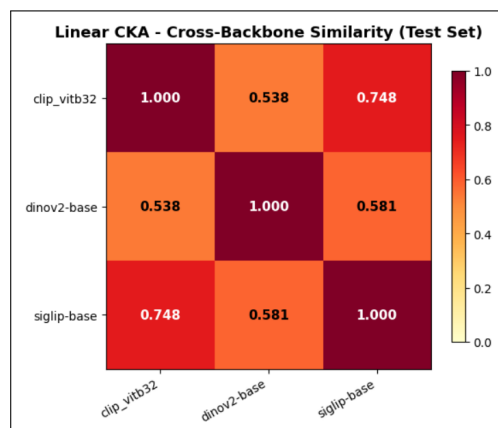


Figure 9: Cross-backbone CKA heatmap. DINOv2 is the most distinct representation.

11 Localization Results

11.1 Baseline Localizer Performance

Table 10: Patch localizer performance on tampered test images (layer L-1).

Backbone	Grid	Mean IoU	Mean Dice	Mean AUC
CLIP ViT-B/32	7×7	0.375	0.504	0.805
DINOv2-Base	16×16	0.323	0.441	0.803
SigLIP-Base	14×14	0.279	0.403	0.771

Surprisingly, CLIP ViT-B/32 achieves the highest localization IoU (0.375) despite having the coarsest spatial resolution (7×7 vs. 16×16 for DINOv2). AUC is uniformly high (≥ 0.77) across all backbones, indicating that the patch-level ranking is accurate; the binary threshold for IoU/Dice penalizes coarse grids less severely when tampered regions are large.

11.2 Localization Stratified by Mask Size

Table 11: IoU stratified by tampered region size (test set).

Backbone	Mask Size	Count	Mean IoU	Mean AUC
CLIP ViT-B/32	Small ($<15\%$)	134	0.237	0.825
	Medium (15–40%)	139	0.445	0.811
	Large ($>40\%$)	60	0.524	0.750
DINOv2-Base	Small	172	0.212	0.810
	Medium	121	0.440	0.821
	Large	40	0.445	0.721
SigLIP-Base	Small	170	0.150	0.777
	Medium	119	0.374	0.779
	Large	44	0.525	0.727

Small tampered regions ($<15\%$ of pixels) are hardest to localize (IoU 0.15–0.24), since few patches fall inside the mask. AUC remains high even for small masks (~ 0.80), confirming that the model correctly *ranks* patches but cannot cleanly *threshold* them when the positive region spans only 1–2 patches.

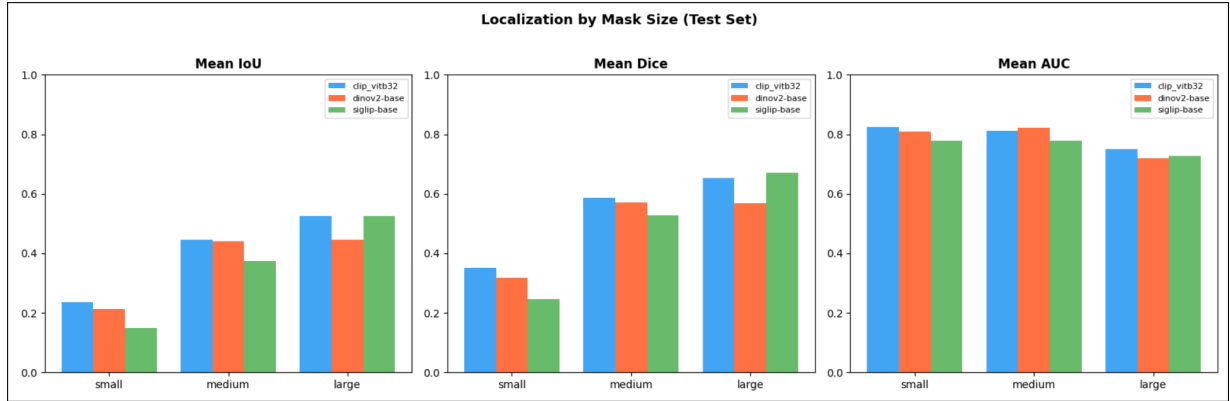


Figure 10: Localization metrics stratified by tampered region size. Small masks are hardest; AUC remains high even when IoU is low.

11.3 Token-CAM: Does the Classifier Attend to Tampered Regions?

Token-CAM projects the classifier’s tampered-class weight vector w_{tamp} onto each patch token: $s_i = w_{\text{tamp}}^\top t_i$. This reveals whether the *classification* decision attends to the actual tampered regions.

Table 12: Token-CAM vs. dedicated patch localizer (test set).

Backbone	Method	IoU	Dice	AUC	Pointing
DINOv2	Token-CAM	0.195	0.297	0.542	0.279
	Localizer	0.323	0.441	0.803	—
SigLIP	Token-CAM	0.201	0.305	0.447	0.213
	Localizer	0.279	0.403	0.771	—
CLIP	Token-CAM	0.241	0.354	0.486	0.228
	Localizer	0.375	0.504	0.805	—

Token-CAM substantially underperforms the dedicated localizer (IoU gap of 0.08–0.13), with pointing accuracy (22–28%) only modestly above chance. This is a central interpretability result: the classifier detects tampering through *holistic* embedding shifts, not by localizing the manipulated region. The same conclusion will reappear in Part II, where SAE feature ablation shows that tampered detection is distributed rather than spatial.

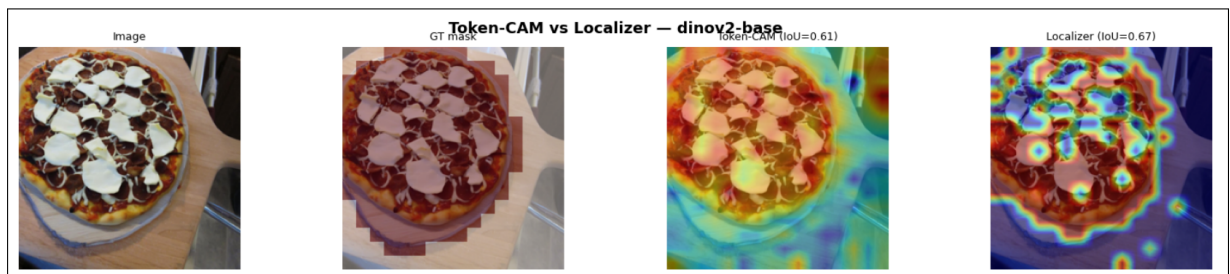


Figure 11: Token-CAM (classifier attention) vs. dedicated localizer. The classifier’s attention is diffuse; the localizer focuses on actual tampered regions.

11.4 Patch Score Distributions and IoU Histograms

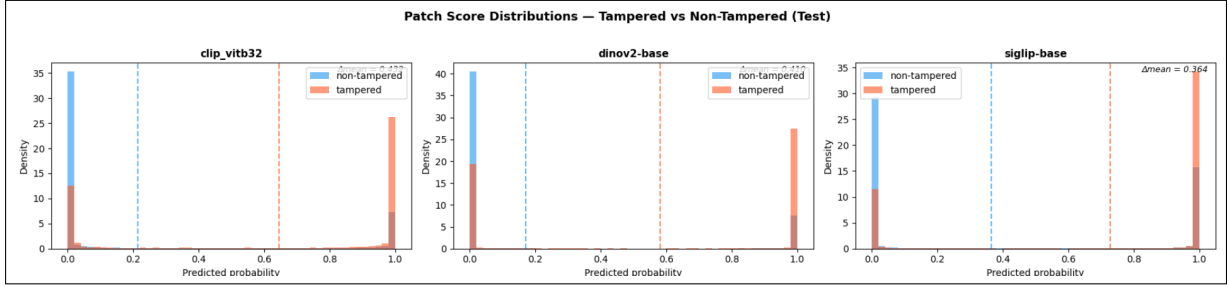


Figure 12: Predicted probability distributions for tampered vs. non-tampered patches. Higher mean separation Δ indicates better discrimination.

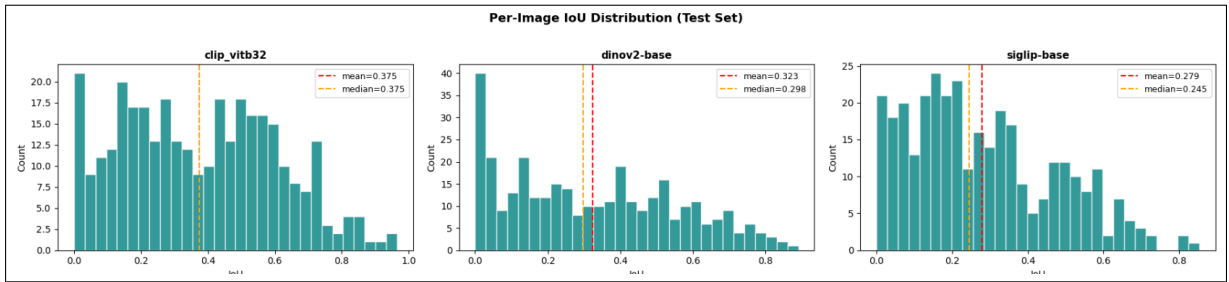


Figure 13: Per-image IoU distributions. CLIP: mean=0.375, DINOv2: mean=0.323, SigLIP: mean=0.279. High variance ($\text{std} \approx 0.2$) indicates inconsistent localization quality.

12 Part I Summary

Part I established that frozen foundation-model features are surprisingly effective for deepfake detection: all three backbones reach 86–90% F1 with a simple linear probe, approaching the full fine-tuned SIDA-7B (93.5%) at a tiny fraction of the cost. The interpretability suite localized the relevant signal to the last quarter of the backbone, showed that it is class-specific (not shortcut-driven), holistic rather than spatial, and robust to social-media degradation. Two findings in particular motivate Part II: (1) tampered detection is markedly harder than real or synthetic, and (2) the classification decision is globally distributed across dimensions. We now turn to the question of how a fine-tuned state-of-the-art detector (SIDA-7B) internally organizes this signal, using sparse autoencoders.

PART II

Mechanistic Interpretation of SIDA via Sparse Autoencoders

Part I characterized the *external* behavior of frozen-feature classifiers on SID-Set. Part II opens the black box of a fine-tuned state-of-the-art detector. We extract the 4096-dimensional classification embedding from SIDA-7B [1] and decompose it with five sparse autoencoder variants. We then perform causal interventions (ablation, amplification) to test which features are *necessary* and *sufficient* for each class decision, analyze the decision boundary by confidence stratification, and map per-class adversarial vulnerability.

13 Background

13.1 SIDA: Deepfake Detection Framework

SIDA [1] is a multimodal framework built on LLaVA [37] and SAM [38] that performs three tasks: deepfake detection (3-class classification), tampered region localization (segmentation), and textual explanation. The architecture processes images through a CLIP vision encoder and a LLaMA-based language model. A special [CLS] token’s last-layer hidden state $h_{\text{det}} \in \mathbb{R}^{4096}$ is passed through a classification head ($\text{Linear}(4096 \rightarrow 2048) \rightarrow \text{ReLU} \rightarrow \text{Linear}(2048 \rightarrow 3)$) to produce detection logits over three classes: real (label 0), fully synthetic (label 1), and tampered (label 2). Our analyses in this part operate directly on h_{det} and the saved head weights.

13.2 Sparse Autoencoders for Mechanistic Interpretability

A sparse autoencoder decomposes a dense activation vector $\mathbf{x} \in \mathbb{R}^d$ into a sparse code $\mathbf{z} \in \mathbb{R}^m$ (with $m \gg d$) such that $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{z} + \mathbf{b}$ approximates $\mathbf{x}$. The overcomplete basis ($m > d$) allows each feature to represent a more monosemantic concept than the original dense dimensions [32]. Sparsity is enforced through various mechanisms: L1 penalties [31], top- k selection [33], gating [34], learnable thresholds (JumpReLU) [35], or nested structure (Matryoshka) [36].

14 Method

14.1 Embedding Extraction

We extract h_{det} from SIDA-7B for 60,000 test images from SID-Set [1] (20,000 per class: real images from OpenImages V7, fully synthetic generated by FLUX, and tampered with local edits via Latent Diffusion). Each embedding is a 4096-dimensional vector stored in float16 precision. We also extract the classification head weights so that, given any modified h_{det} , we can compute classification logits without re-running the full model.

14.2 SAE Training Setup

We split the 60,000 embeddings into 30,000 for SAE training and 30,000 for evaluation, using stratified sampling to preserve equal class balance (10,000 per class in each split). Embeddings are normalized to zero mean and unit variance per feature dimension using statistics computed on the training split. We train five SAE variants, all with input dimension $d = 4096$ and hidden dimension $m = 16,384$ ($4\times$ expansion):

1. **Vanilla SAE** [31]: ReLU activations with an L1 sparsity penalty ($\lambda = 5 \times 10^{-4}$). Loss: $\mathcal{L} = \text{MSE}(\hat{\mathbf{x}}, \mathbf{x}) + \lambda \|\mathbf{z}\|_1$.
2. **TopK SAE** [33]: Enforces exact sparsity by retaining only the top $k = 64$ activations per sample. No L1 penalty needed since sparsity is structural. Loss: $\mathcal{L} = \text{MSE}(\hat{\mathbf{x}}, \mathbf{x})$.
3. **Gated SAE** [34]: A sigmoid gating network decides *whether* a feature is active; a separate magnitude network decides *how much*. Activation: $z_i = \sigma(\mathbf{w}_{\text{gate},i}^\top \mathbf{x}) \cdot \text{ReLU}(\mathbf{w}_{\text{mag},i}^\top \mathbf{x}) \cdot e^{r_i}$. L1 penalty ($\lambda = 5 \times 10^{-4}$) on gate pre-activations.
4. **JumpReLU SAE** [35]: Per-feature learnable threshold θ_i , activations below θ_i zeroed: $z_i = \text{ReLU}(a_i) \cdot \mathbb{I}[a_i > \theta_i]$. Straight-through estimator for gradient flow through the threshold. L0-approximate penalty ($\lambda = 5 \times 10^{-4}$): $\sum_i \sigma(10(a_i - \theta_i))$.
5. **Matryoshka SAE** [36]: Features are ordered by importance. Training applies the reconstruction loss at nested widths $\{2048, 4096, 8192, 16384\}$: the first w features form a valid sub-SAE for each width w . Total loss averages reconstruction + L1 across all nesting levels.

All models are trained for 50 epochs with Adam ($\text{lr} = 3 \times 10^{-4}$), cosine annealing, batch size 256, and gradient clipping at 1.0. Decoder weight columns are constrained to unit ℓ_2 norm after each step, following standard practice [31].

14.3 Analysis Framework

Feature selectivity. For each SAE feature i and class c , we compute the mean activation $\mu_{i,c} = \mathbb{E}[z_i \mid y = c]$. The *selectivity score* for feature i with respect to class c is

$$S_{i,c} = \frac{\mu_{i,c} - \bar{\mu}_{i,-c}}{\mu_{i,c} + \bar{\mu}_{i,-c} + \epsilon}, \quad (1)$$

where $\bar{\mu}_{i,-c}$ is the mean activation across the other two classes. A selectivity of 1.0 means the feature activates exclusively for class c ; 0.0 means equal activation across classes.

Intervention framework. Given a trained SAE, we perform causal interventions by modifying the sparse code $\mathbf{z}$ before decoding. **Ablation:** $z_i \leftarrow 0$ for target features $i \in \mathcal{S}$, testing necessity. **Amplification:** $z_i \leftarrow \alpha \cdot z_i$ for target features, testing whether strengthening the feature changes classification. The modified code is decoded, denormalized, and passed through the saved classification head to produce logits. The change in accuracy quantifies the causal effect.

Decision boundary analysis. For each class we identify “hard” samples (bottom 20th percentile of softmax confidence for the true class) and “easy” samples (top 20th percentile). For each feature we compute the standardized mean difference between groups:

$$d_i = \frac{\bar{z}_{i,\text{easy}} - \bar{z}_{i,\text{hard}}}{\sigma_{i,\text{pooled}}}. \quad (2)$$

Features with large $|d_i|$ are “boundary features”: their activation level determines whether the model correctly classifies ambiguous samples.

Adversarial vulnerability. We define a per-feature *vulnerability score* as

$$V_{i,c} = \frac{|A_{i,c}|}{\sigma_{z_i}}, \quad (3)$$

where $A_{i,c}$ is the linear logit attribution of feature i to class c (computed by projecting the decoder direction through the classification head weights) and σ_{z_i} is the activation standard deviation. Features with high V have large classification impact but low variance, so small perturbations to them disproportionately affect the output.

15 Results

15.1 SAE Reconstruction Quality

Table 13 summarizes the five SAE variants on reconstruction fidelity and sparsity metrics. *Reconstruction MSE* is $\frac{1}{n} \sum_j \|\hat{\mathbf{x}}_j - \mathbf{x}_j\|^2$ on 5,000 test samples. *Explained variance* is $1 - \text{MSE}/\text{Var}(\mathbf{x})$. *Active features* counts features that fire for at least one test sample. *Features per sample* is the mean number of nonzero activations per input. *Dead features* are features that never activate, representing wasted capacity.

Table 13: SAE reconstruction and sparsity comparison. All models use $d_{\text{hidden}} = 16,384$ ($4\times$ expansion). Metrics computed on 5,000 test samples.

SAE	Recon MSE	Exp. Var.	Active Feats	Feats/Sample	Dead %
Vanilla	0.0328	96.7%	16,384	9,556	0.0%
TopK	0.2812	71.9%	6,047	64	63.1%
Gated	0.0419	95.8%	16,384	8,966	0.0%
JumpReLU	0.0392	96.1%	16,384	9,233	0.0%
Matryoshka	0.0379	96.2%	16,384	9,387	0.0%

The Vanilla SAE achieves the best reconstruction (MSE 0.0328, 96.7% explained variance). The TopK SAE, constrained to exactly 64 active features per sample (0.39% of the 16,384 hidden dimensions), sacrifices significant reconstruction quality (71.9% EV) and exhibits 63.1% dead features: 10,337 of its 16,384 features never activate. All other variants utilize their full capacity. Among the non-TopK variants, Matryoshka (96.2%) and JumpReLU (96.1%) are competitive with Vanilla, while Gated lags slightly at 95.8%.

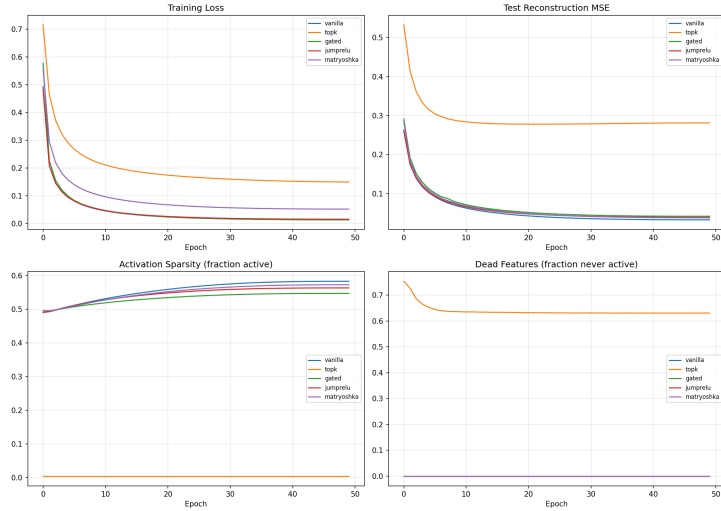


Figure 14: Training dynamics over 50 epochs. TopK (orange) converges slowly and plateaus at high reconstruction error with 63% dead features. All other variants converge by epoch 30 with 0% dead features.

15.2 Classification Accuracy Through the SAE

A key question is whether the SAE preserves the information needed for classification. We pass test embeddings through the full SAE pipeline (encode $\rightarrow$ sparse code $\rightarrow$ decode) then through the original classification head, and compare to the original accuracy (embeddings straight to head, no SAE).

Table 14: Classification accuracy when embeddings pass through SAE (encode $\rightarrow$ decode $\rightarrow$ cls_head) vs. the original direct path. All values are percentages.

Method	Overall	Real	Full Synthetic	Tampered
Original (no SAE)	93.60	98.39	99.92	82.50
Vanilla	93.60	98.39	99.92	82.50
Gated	93.62	98.38	99.93	82.55
Matryoshka	93.58	98.32	99.92	82.51
JumpReLU	93.57	98.42	99.92	82.38
TopK	92.73	98.10	99.82	80.26

Vanilla, Gated, JumpReLU, and Matryoshka preserve classification accuracy within 0.05 percentage points of the original, meaning the SAE reconstruction introduces negligible information loss for the downstream task. The TopK SAE loses 2.24 percentage points on tampered images (80.26% vs. 82.50%), confirming that 64 features are insufficient to fully represent the tampered-class signal. The per-class pattern here echoes exactly what Part I found: **fully synthetic is classified near-perfectly (99.82–99.93%)**, while **tampered lags at 80–82%**.

15.3 Feature Selectivity Analysis

Table 15: Mean selectivity score of top-20 discriminative features per class. Higher is more class-specific. Computed on 30,000 test samples.

SAE	Real	Full Synthetic	Tampered
Vanilla	0.476	0.787	0.515
TopK	0.959	1.000	0.959
Gated	0.483	0.867	0.524
JumpReLU	0.547	0.893	0.595
Matryoshka	0.446	0.916	0.494

Three patterns emerge. First, **Full Synthetic features are the most selective** across all SAE types (0.787–1.000): the model develops highly specialized features for detecting FLUX-generated content. This is the SAE-level mirror of the Part I finding that synthetic-class L1 coefficients are dramatically larger than real/tampered. Second, **Real and Tampered features are less selective** (0.446–0.595), indicating more representational overlap, again consistent with Part I’s PCA/t-SNE results. Third, the **TopK SAE achieves near-perfect selectivity** (0.959–1.000) because its extreme sparsity constraint (64 features) forces each feature to specialize for a single class.

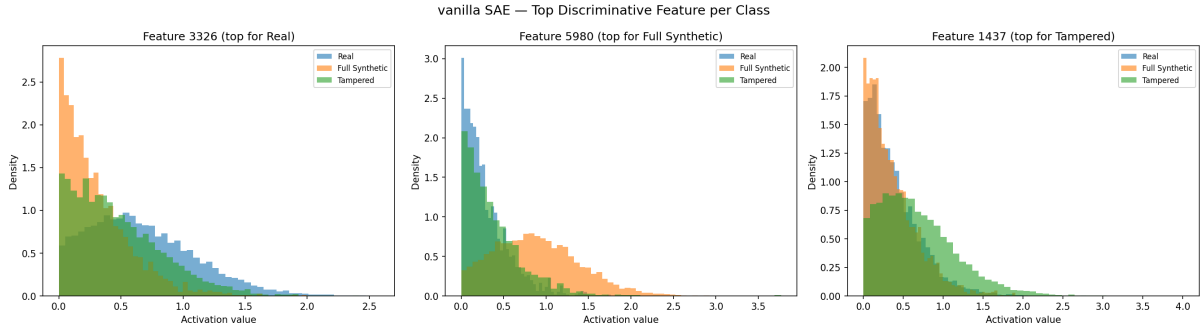


Figure 15: Activation distributions of the top discriminative feature per class (Vanilla SAE). **Left:** Feature 3326 (Real) activates 0–2.5 for real images, near-zero for fake. **Center:** Feature 5980 (Full Synthetic) has a bimodal distribution extending to 3.5, highly selective. **Right:** Feature 1437 (Tampered) shows less clean separation; tampered activations extend to 4.0 but overlap with real.

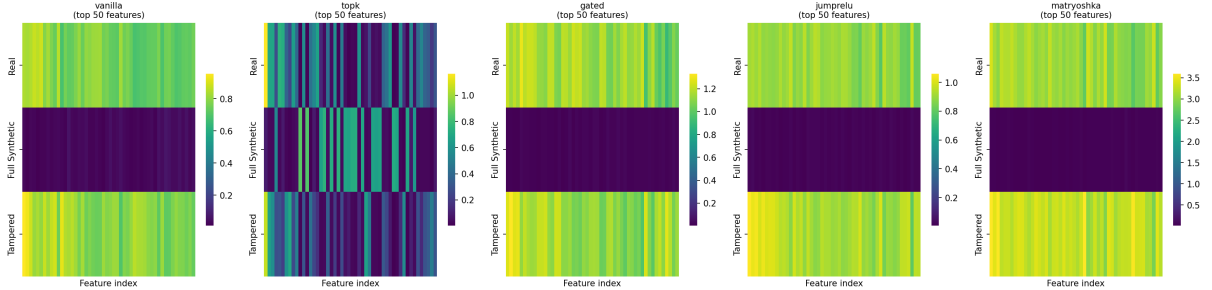


Figure 16: Per-class mean activation of the top 50 features for each SAE. Each row is a class (Real, Full Synthetic, Tampered), each column a feature. The TopK SAE shows extreme sparsity; others show dense but class-dependent magnitude patterns.

15.4 Per-Class Activation Statistics

Table 16: Mean number of active features per sample, by class. Computed on 30,000 test samples.

SAE	Overall	Real	Full Synthetic	Tampered
Vanilla	9,555	9,654	9,436	9,573
TopK	64	64	64	64
Gated	8,964	9,301	8,609	8,980
JumpReLU	6,756	6,626	6,856	6,786
Matryoshka	9,384	9,639	9,020	9,493

The JumpReLU SAE achieves notably higher sparsity (6,756 features per sample vs. 9,555 for Vanilla) while maintaining comparable reconstruction quality, due to its learnable per-feature thresholds. In Vanilla, Gated, and Matryoshka, real images activate slightly *more* features than synthetic or tampered, suggesting the detector encodes authentic visual content with richer representations.

15.5 Feature Ablation Study

We ablate (set to zero) the top- K most discriminative features for each target class and measure the effect on all classes’ accuracy. Tables 17–19 present the complete results for all five SAEs.

Table 17: Ablation of top- K **Real** discriminative features. Δ All shows change in overall accuracy from SAE baseline.

SAE	K	Overall (Δ All)	Real	Synthetic	Tampered
Vanilla (93.60)	5	93.70 (+0.10)	98.33	99.92	82.84
	10	93.78 (+0.18)	98.28	99.93	83.12
	20	93.87 (+0.27)	98.09	99.93	83.58
TopK (92.73)	5	92.73 (+0.00)	98.10	99.82	80.26
	10	92.73 (+0.00)	98.09	99.82	80.27
	20	92.73 (+0.00)	98.07	99.83	80.28
Gated (93.62)	5	93.71 (+0.09)	98.34	99.93	82.85
	10	93.75 (+0.13)	98.23	99.93	83.10
	20	93.90 (+0.28)	98.13	99.94	83.63
JumpReLU (93.57)	5	93.75 (+0.18)	98.39	99.92	82.95
	10	93.79 (+0.22)	98.37	99.92	83.08
	20	93.88 (+0.31)	98.30	99.92	83.41
Matryoshka (93.58)	5	94.00 (+0.42)	98.04	99.93	84.03
	10	94.63 (+1.05)	97.37	99.94	86.58
	20	95.22 (+1.64)	95.62	99.97	90.08

Table 18: Ablation of top- K **Full Synthetic** discriminative features. Synthetic accuracy is virtually immune to ablation across all SAEs.

SAE	K	Overall (Δ All)	Real	Synthetic	Tampered
Vanilla (93.60)	5	93.60 (+0.00)	98.39	99.91	82.51
	10	93.61 (+0.01)	98.40	99.91	82.53
	20	93.62 (+0.02)	98.42	99.90	82.54
TopK (92.73)	5	92.73 (+0.00)	98.10	99.82	80.26
	10	92.73 (+0.00)	98.10	99.82	80.26
	20	92.73 (+0.00)	98.10	99.82	80.26
Gated (93.62)	5	93.62 (+0.00)	98.39	99.92	82.56
	10	93.63 (+0.01)	98.39	99.92	82.57
	20	93.64 (+0.02)	98.41	99.91	82.59
JumpReLU (93.57)	5	93.67 (+0.10)	98.42	99.92	82.68
	10	93.68 (+0.11)	98.41	99.92	82.71
	20	93.68 (+0.11)	98.41	99.92	82.72
Matryoshka (93.58)	5	93.61 (+0.03)	98.37	99.91	82.56
	10	93.62 (+0.04)	98.39	99.90	82.57
	20	93.67 (+0.09)	98.50	99.90	82.60

Table 19: Ablation of top- K **Tampered** discriminative features. The Matryoshka SAE shows catastrophic overall accuracy loss at $K = 20$.

SAE	K	Overall (Δ All)	Real	Synthetic	Tampered
Vanilla (93.60)	5	93.44 (−0.16)	98.42	99.92	81.98
	10	93.31 (−0.29)	98.46	99.92	81.56
	20	93.04 (−0.56)	98.52	99.92	80.68
TopK (92.73)	5	92.72 (−0.01)	98.10	99.82	80.25
	10	92.70 (−0.03)	98.11	99.82	80.18
	20	92.62 (−0.11)	98.11	99.82	79.92
Gated (93.62)	5	93.45 (−0.17)	98.38	99.93	82.05
	10	93.32 (−0.30)	98.39	99.93	81.64
	20	93.09 (−0.53)	98.42	99.93	80.92
JumpReLU (93.57)	5	93.55 (−0.02)	98.42	99.92	82.31
	10	93.43 (−0.14)	98.43	99.92	81.93
	20	93.25 (−0.32)	98.49	99.92	81.33
Matryoshka (93.58)	5	92.16 (−1.42)	98.58	99.91	78.00
	10	90.54 (−3.04)	98.70	99.91	73.00
	20	86.64 (−6.94)	98.89	99.91	61.13

Five key findings emerge from the ablation study. (1) **Ablating Real features improves Tampered accuracy** across all SAEs: removing the top-20 Real features for Vanilla raises tampered accuracy from 82.50% to 83.58% (+1.08pp), and for Matryoshka the effect is dramatic (+7.57pp to 90.08%). This suggests some real-discriminative features actively suppress tampered detection. (2) **Ablating Tampered features hurts Tampered accuracy**, confirming these features are causally necessary. The effect is strongest for Matryoshka (−21.38pp at $K = 20$, from 82.51% to 61.13%). (3) **Synthetic accuracy is immune to all ablations**: it never drops below 99.82% regardless of which features are removed, for any SAE. The synthetic detection signal is distributed broadly and redundantly, a striking echo of Part I’s finding that the L1-synthetic coefficients are concentrated yet the overall synthetic classification is robust. (4) **Effects saturate at $K = 20$** : $K = 50$ and $K = 100$ produce identical results for all SAEs, meaning the top 20 features carry the full class-discriminative signal. (5) **TopK SAE is immune to ablation**: with only 64 active features, the discriminative features identified by selectivity are rarely the same ones active for a given sample.

15.6 Feature Steering

We amplify the top-20 discriminative features per class by various scale factors and measure the effect. Table 20 shows steering toward tampered detection (the weakest class).

Table 20: Feature steering toward **Tampered**: amplifying top-20 tampered-discriminative features. ΔT = change in Tampered accuracy, ΔAll = change in overall accuracy from SAE baseline.

SAE	Scale	Overall (ΔAll)	Real	Synthetic	Tampered	ΔT
Vanilla	1.5 $\times$	93.85 (+0.25)	98.30	99.92	83.33	+0.83
	3.0 $\times$	94.41 (+0.81)	97.96	99.91	85.37	+2.87
	5.0 $\times$	95.03 (+1.43)	97.43	99.91	87.74	+5.24
	10.0 $\times$	95.49 (+1.89)	94.67	99.89	91.92	+9.42
TopK	1.5 $\times$	92.77 (+0.04)	98.10	99.82	80.40	+0.14
	3.0 $\times$	92.85 (+0.12)	98.04	99.82	80.70	+0.44
	5.0 $\times$	92.92 (+0.19)	97.98	99.82	80.97	+0.71
	10.0 $\times$	92.95 (+0.22)	97.76	99.82	81.27	+1.01
Gated	1.5 $\times$	93.81 (+0.19)	98.32	99.92	83.18	+0.63
	3.0 $\times$	94.28 (+0.66)	98.04	99.92	84.88	+2.33
	5.0 $\times$	94.86 (+1.24)	97.64	99.91	87.04	+4.49
	10.0 $\times$	95.54 (+1.92)	95.52	99.89	91.20	+8.65
JumpReLU	1.5 $\times$	93.83 (+0.26)	98.37	99.92	83.19	+0.81
	3.0 $\times$	94.17 (+0.60)	98.03	99.92	84.55	+2.17
	5.0 $\times$	94.65 (+1.08)	97.63	99.92	86.41	+4.03
	10.0 $\times$	95.39 (+1.82)	96.43	99.91	89.84	+7.46
Matryoshka	1.5 $\times$	95.08 (+1.50)	97.08	99.93	88.22	+5.71
	3.0 $\times$	92.80 (−0.78)	82.21	99.93	96.25	+13.74
	5.0 $\times$	79.63 (−13.95)	39.86	99.91	99.13	+16.62
	10.0 $\times$	67.67 (−25.91)	3.58	99.72	99.72	+17.21

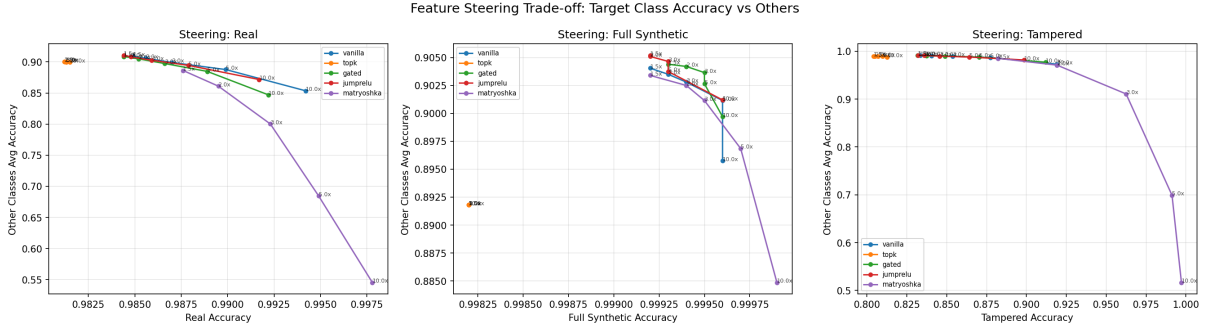


Figure 17: Trade-off between target-class accuracy and other classes' mean accuracy under feature steering at scales 1.5 $\times$ to 10 $\times$. Each point is annotated with the scale factor. The Matryoshka SAE shows the most extreme trade-offs.

Tables 21 and 22 show steering toward real and synthetic detection.

Table 21: Feature steering toward **Real**: amplifying top-20 real-discriminative features. ΔR = change in Real accuracy, ΔAll = change in overall accuracy.

SAE	Scale	Overall (ΔAll)	Real	Synthetic	Tampered	ΔR
Vanilla	1.5 $\times$	93.44 (−0.16)	98.49	99.90	81.93	+0.10
	3.0 $\times$	92.84 (−0.76)	98.70	99.89	79.93	+0.31
	5.0 $\times$	92.18 (−1.42)	98.99	99.89	77.65	+0.60
	10.0 $\times$	90.05 (−3.55)	99.42	99.87	70.85	+1.03
TopK	1.5 $\times$	92.73 (+0.00)	98.12	99.82	80.25	+0.02
	3.0 $\times$	92.73 (+0.00)	98.13	99.82	80.25	+0.03
	5.0 $\times$	92.71 (−0.02)	98.13	99.80	80.20	+0.03
	10.0 $\times$	92.69 (−0.04)	98.15	99.79	80.12	+0.05
Gated	1.5 $\times$	93.39 (−0.23)	98.44	99.92	81.81	+0.06
	3.0 $\times$	92.72 (−0.90)	98.66	99.90	79.59	+0.28
	5.0 $\times$	91.95 (−1.67)	98.89	99.89	77.06	+0.51
	10.0 $\times$	89.54 (−4.08)	99.22	99.88	69.52	+0.84
JumpReLU	1.5 $\times$	93.52 (−0.05)	98.44	99.92	82.21	+0.02
	3.0 $\times$	93.04 (−0.53)	98.58	99.91	80.64	+0.16
	5.0 $\times$	92.55 (−1.02)	98.79	99.90	78.97	+0.37
	10.0 $\times$	91.17 (−2.40)	99.17	99.87	74.47	+0.75
Matryoshka	1.5 $\times$	92.00 (−1.58)	98.76	99.90	77.34	+0.44
	3.0 $\times$	86.43 (−7.15)	99.23	99.89	60.16	+0.91
	5.0 $\times$	78.81 (−14.77)	99.49	99.84	37.10	+1.17
	10.0 $\times$	69.61 (−23.97)	99.78	99.47	9.58	+1.46

Table 22: Feature steering toward **Full Synthetic**: amplifying top-20 synthetic-discriminative features. ΔS = change in Synthetic accuracy, ΔAll = change in overall accuracy.

SAE	Scale	Overall (ΔAll)	Real	Synthetic	Tampered	ΔS
Vanilla	1.5 $\times$	93.58 (−0.02)	98.35	99.92	82.46	+0.00
	3.0 $\times$	93.50 (−0.10)	98.18	99.94	82.38	+0.02
	5.0 $\times$	93.40 (−0.20)	98.01	99.96	82.22	+0.04
	10.0 $\times$	93.04 (−0.56)	97.51	99.96	81.64	+0.04
TopK	1.5 $\times$	92.73 (+0.00)	98.10	99.82	80.26	+0.00
	3.0 $\times$	92.73 (+0.00)	98.10	99.82	80.26	+0.00
	5.0 $\times$	92.73 (+0.00)	98.10	99.82	80.26	+0.00
	10.0 $\times$	92.73 (+0.00)	98.10	99.82	80.26	+0.00
Gated	1.5 $\times$	93.60 (−0.02)	98.35	99.93	82.53	+0.00
	3.0 $\times$	93.56 (−0.06)	98.29	99.95	82.44	+0.02
	5.0 $\times$	93.49 (−0.13)	98.24	99.95	82.29	+0.02
	10.0 $\times$	93.30 (−0.32)	97.94	99.96	82.00	+0.03
JumpReLU	1.5 $\times$	93.65 (+0.08)	98.40	99.92	82.63	+0.00
	3.0 $\times$	93.62 (+0.05)	98.36	99.93	82.57	+0.01
	5.0 $\times$	93.56 (−0.01)	98.26	99.93	82.49	+0.01
	10.0 $\times$	93.40 (−0.17)	97.96	99.96	82.28	+0.04
Matryoshka	1.5 $\times$	93.53 (−0.05)	98.20	99.91	82.48	−0.01
	3.0 $\times$	93.39 (−0.19)	97.97	99.95	82.26	+0.03
	5.0 $\times$	93.11 (−0.47)	97.46	99.97	81.91	+0.05
	10.0 $\times$	92.32 (−1.26)	96.17	99.99	80.80	+0.07

The steering results demonstrate **causal control** over the classifier. For the Vanilla SAE,

amplifying tampered features by $10\times$ improves tampered accuracy from 82.50% to 91.92% (+9.42pp), the largest per-class improvement, at the cost of 3.72pp on real accuracy. The Matryoshka SAE shows extreme steering: at $10\times$ tampered amplification, tampered accuracy reaches 99.72% but real accuracy collapses to 3.58%, indicating its features are highly entangled. TopK SAE is nearly immune to steering (tampered improves by only 1.01pp at $10\times$), consistent with its poor reconstruction quality. Synthetic detection cannot be meaningfully improved by steering because it is already saturated at 99.9%; this, too, aligns with the Part I observation that synthetic detection rides on a concentrated, pervasive signal.

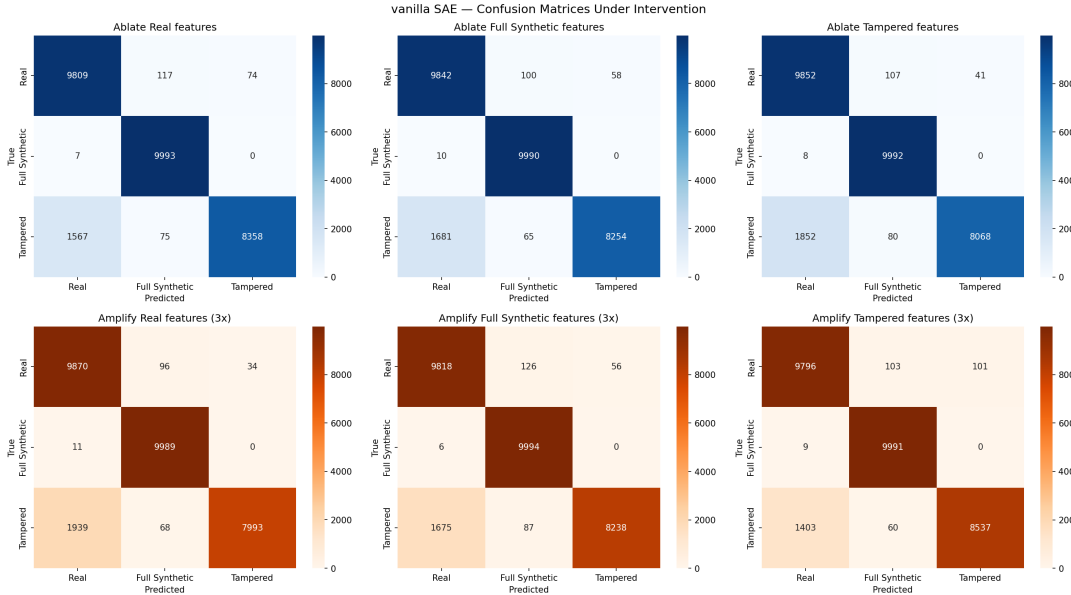


Figure 18: Confusion matrices for the Vanilla SAE under ablation (top row) and amplification (bottom row) of class-discriminative features. Notable: ablating Synthetic features causes 3,481 Tampered→Real misclassifications; amplifying Tampered features improves Tampered recall from 8,068 to 8,537.

15.7 Decision Boundary Analysis

We analyze “hard” samples (bottom 20% of classification confidence) vs. “easy” samples (top 20%) per class.

Table 23: Decision boundary analysis. Hard = bottom 20% confidence, Easy = top 20%. “Conf.” = mean softmax probability for the true class. “Top Feature” = SAE feature with largest standardized mean difference between hard and easy samples.

Class	n_{hard}	Hard Acc.	Hard Conf.	Easy Acc.	Easy Conf.	Top Feature (effect)
Real	2,000	91.95%	0.865	100%	0.999	8532 ($d = 0.88$)
Full Synth.	1,193	99.33%	0.992	n/a	n/a	n/a
Tampered	2,000	12.70%	0.210	100%	0.999	8865 ($d = 1.10$)

The most critical finding: **hard tampered samples achieve only 12.7% accuracy** with a mean confidence of 0.21, barely above random guessing among three classes. This represents a severe failure mode: approximately 4,000 tampered images (20% of the tampered test set) are effectively undetectable. In contrast, hard real images still achieve 91.95% accuracy (confidence 0.87), and hard synthetic images achieve 99.33% even at their lowest confidence (0.99).

For Full Synthetic, no samples qualify as “easy” (top 20% confidence) because the model

classifies virtually all synthetic images with near-identical high confidence: there is no confidence gradient to separate hard from easy. This is the mechanistic counterpart to the Part I observation that synthetic is the easiest class by F1 across all three backbones.

Feature 8865 shows the strongest boundary discrimination for tampered images (effect size $d = 1.10$): easy tampered samples have significantly higher activation values (1.0–2.5) compared to hard tampered samples (0–1.0), as shown in Figure 19. This feature could serve as a confidence calibration signal.

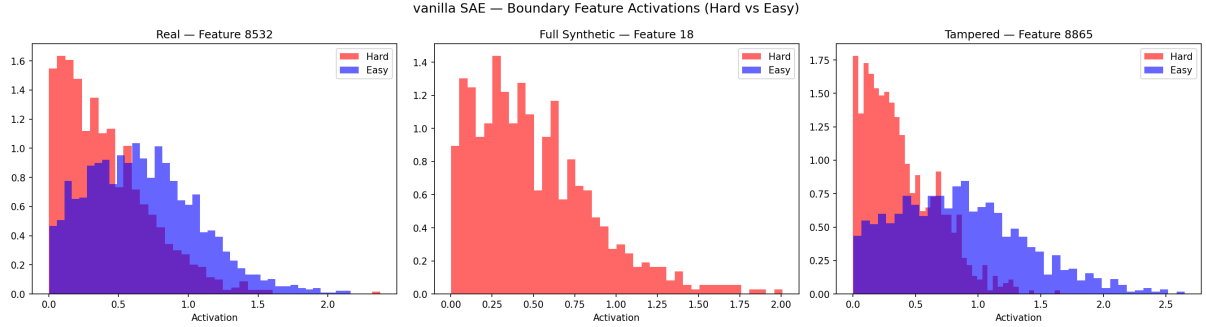


Figure 19: Activation distributions of the top boundary feature for hard (red) vs. easy (blue) samples per class. **Right:** Feature 8865 for Tampered shows clear separation; hard samples (12.7% accuracy) have much lower activations.

15.8 Adversarial Vulnerability Mapping

Table 24: Top-5 most vulnerable features per class (Vanilla SAE). Vulnerability score $V = |\text{attribution}|/\text{std}$: features with high classification impact but low activation variance are easiest to exploit. Computed on 30,000 test samples.

Class	Feature	Vulnerability Score
Real	5806	0.149
	1955	0.144
	14731	0.140
	8172	0.139
	6233	0.139
Full Synthetic	9211	0.196
	8632	0.192
	6199	0.190
	4825	0.187
	8292	0.186
Tampered	6019	0.188
	4275	0.182
	6636	0.171
	8641	0.170
	15304	0.168

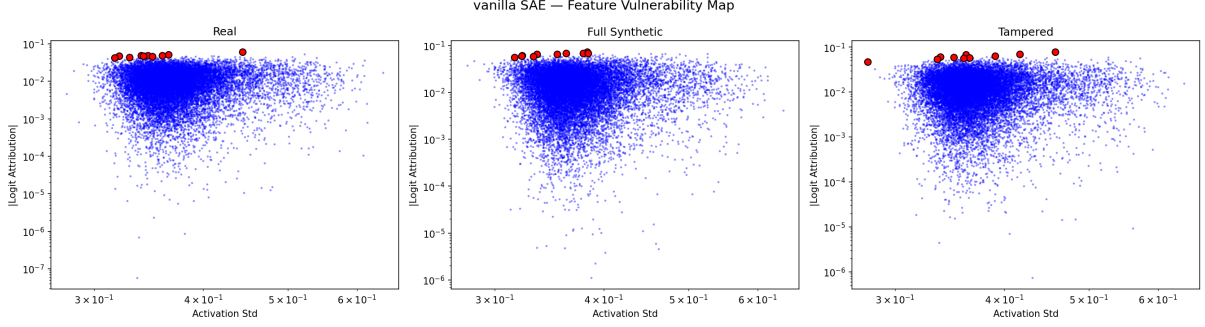


Figure 20: Feature vulnerability map: logit attribution magnitude vs. activation standard deviation (log-log scale). Red points mark the top-10 most vulnerable features. Vulnerable features occupy the upper region (high attribution) with moderate-to-low std.

The fully synthetic class has the highest vulnerability scores (top 0.196 vs. 0.149 for real), meaning its classification relies on features with high logit impact but relatively low activation variance. **Paradoxically, synthetic classification is the most accurate (99.9%) yet also the most fragile to targeted feature perturbation.** While the model has discovered strong synthetic-detection features, an adversary who can perturb these specific directions could evade detection with minimal image modification. Part I’s robustness results showed the *distributional* robustness of these features under benign degradation; this result shows their *adversarial* fragility under targeted attack, and the contrast is informative.

15.9 Logit Attribution

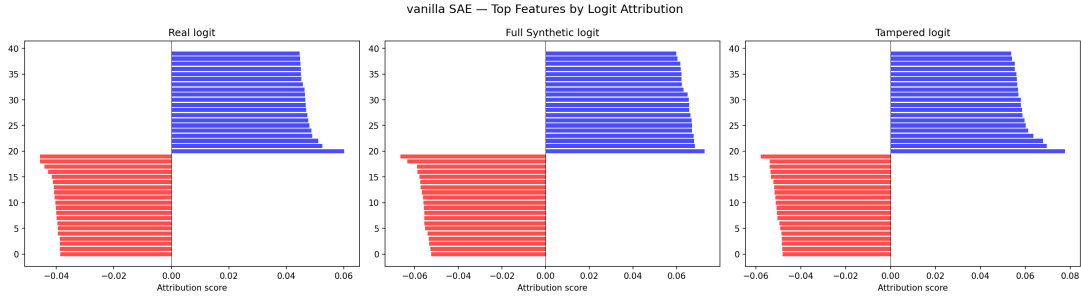


Figure 21: Linear logit attribution of SAE features to each class. Features positively attributing to one logit tend to negatively attribute to others, consistent with softmax competition. Computed as: decoder direction $\times$ cls_head first-layer weights $\times$ cls_head second-layer weights.

The logit attribution decomposes each class logit into per-feature contributions via the linear approximation $A_{i,c} = \mathbf{d}_i^\top W_1^\top W_2^\top$, where $\mathbf{d}_i$ is the i -th decoder direction (denormalized) and W_1, W_2 are the classification head weight matrices. Features that positively push toward one class logit tend to negatively push against the others, reflecting the softmax competition structure.

15.10 UMAP Visualization

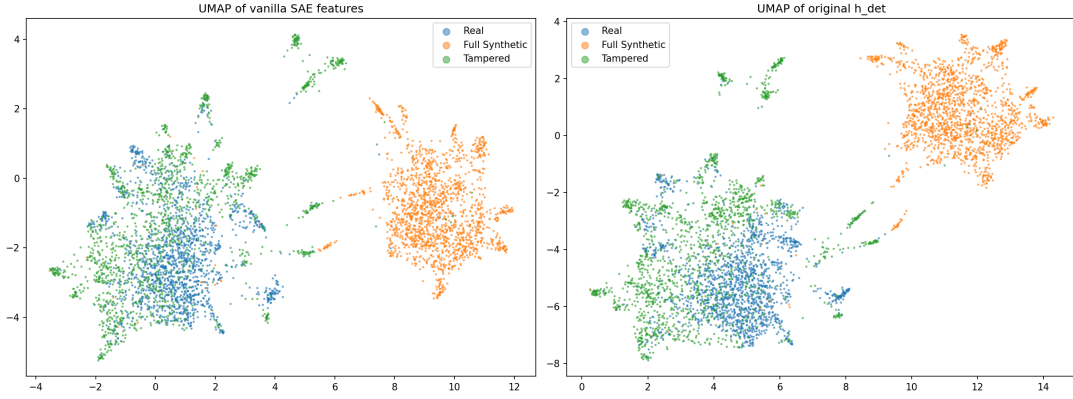


Figure 22: UMAP of SAE features (left) vs. original h_{det} (right) for 5,000 test samples. Both spaces show clear separation of fully synthetic images. Tampered images overlap more with real in both spaces, consistent with the lower classification accuracy, and mirroring the PCA/t-SNE picture from Part I.

Synthesis

16 Discussion: Bridging the Two Analyses

The two parts of this report were conducted on the same benchmark but with very different methodological lenses: Part I trains a linear probe on *frozen* 512–768-d representations, while Part II decomposes the *fine-tuned* 4096-d embedding of a 7B-parameter model into 16,384 sparse features. It is therefore striking how closely the two converge. We highlight four bridges.

Tampered is the hard class, universally. Part I found the worst classification performance on the tampered class across all three frozen backbones (see Table 3 and Figure 1), and the worst localization IoU for small tampered regions. Part II found the same gap in SIDA-7B: 82.5% tampered accuracy vs. 98.4% real and 99.9% synthetic (Table 14). Part II then sharpens the picture with the boundary analysis (Table 23): the tampered difficulty is not a gradual degradation but a *cliff*: the hardest 20% collapse to 12.7% accuracy at confidence 0.21. This is consistent across model scales and pretraining paradigms, suggesting the difficulty is intrinsic to the tampered class rather than an artifact of any single architecture. The underlying reason is the same in both analyses: tampered images occupy a heterogeneous region of the embedding space (many possible local edits, each small) that overlaps with real, whereas synthetic images inhabit a concentrated region shaped by the generator’s pervasive signature.

Synthetic detection uses a concentrated, high-amplitude signal. In Part I, the L1 synthetic-class coefficients are dramatically larger than real or tampered (e.g., +41.13 for DI-NOv2 vs. ± 9.8 for CLIP real). In Part II, the synthetic class has the highest SAE feature selectivity (0.787–1.000 across variants) and is essentially immune to ablation of any 20 features (accuracy never drops below 99.82%). Both analyses paint the same picture: synthetic classification rides on a broadly distributed but concentrated feature signature that is easy to detect but, as the vulnerability map shows, also the most adversarially fragile.

Detection is holistic, not spatial. Part I’s Token-CAM analysis showed that the classifier’s attention is diffuse, not localized on the actual tampered region (pointing accuracy only 22–28%). Part II’s feature ablation shows the same thing from a different angle: tampered classification is distributed across many features, not anchored to a spatially coherent set; and the “top 20 most discriminative” features do not correspond to a small, specialized circuit (ablating them only modestly harms tampered accuracy for Vanilla/Gated/JumpReLU, at most -0.56pp). In both studies, the decision is a holistic embedding shift rather than a localized visual cue.

Class-specific, not shortcut. The SHAP shortcut analysis in Part I showed that only 2–3 of the top-20 dimensions are shared across all three classes. The SAE analysis in Part II shows the analogous property: per-class discriminative features are largely distinct, and steering toward one class demonstrably disrupts the others (Tables 20–22). Both studies thus argue against shortcut learning: the detectors use genuine per-class cues, not a shared spurious feature, which is reassuring for deployment.

Where the two studies diverge. Part I emphasizes that frozen features are *robust* to common social-media transforms, while Part II emphasizes that fine-tuned SIDA features are *adversarially fragile*. These are not contradictory: one measures stability under distributional corruption, the other under targeted attack. Together they argue for a deployment picture in which the feature directions that make synthetic detection easy also make it attackable, and that robustness and vulnerability are two sides of the same concentrated feature signature.

SAE variant recommendations. Across the five SAE architectures, the Vanilla SAE offers the best overall trade-off: lowest reconstruction error, no accuracy loss through the encode/decode cycle, and moderate feature selectivity. The JumpReLU SAE offers higher sparsity (6,756 active features per sample vs. 9,556 for Vanilla) with minimal accuracy cost, making it attractive when sparsity is the priority. The Matryoshka SAE is uniquely suited for multi-scale analysis but its features are more entangled, causing the dramatic steering behavior in Tables 20–21. The TopK SAE, despite its interpretably sparse codes (64 features), loses too much information for faithful reconstruction and is immune to intervention.

Implications for adversarial robustness. The Part II vulnerability map provides a concrete target list: the top-20 vulnerable features per class identify the exact directions an adversary would perturb. Adversarial training that specifically targets these directions could plausibly improve robustness without sacrificing accuracy. The Part I robustness experiments suggest that frozen features are already resistant to benign degradation; combining the two, a promising direction is to defend the fine-tuned model’s vulnerable directions while preserving the frozen-feature regime’s distributional robustness.

17 Conclusion

We presented a two-part interpretability study of deepfake detection on SID-Set. Part I showed that *frozen* features from three foundation models reach 86–90% F1 for classification and 0.375 IoU for localization with no fine-tuning and no GPU at evaluation time, and it identified where and how the pretrained signal exists through eight complementary analyses. Part II showed, via five sparse autoencoder variants, that the *fine-tuned* SIDA-7B model organizes the same signal into causally steerable features: amplifying the top-20 tampered-discriminative features improves tampered accuracy by 9.4 percentage points; ablating tampered features hurts tampered accuracy; synthetic classification is broadly robust yet adversarially the most fragile; and hard tampered images occupy a sharp cliff of 12.7% accuracy.

The two studies converge on a consistent account: tampered images are the hardest class, synthetic images are detected by a concentrated high-amplitude signature, and the decision is holistic rather than spatial in both the frozen and fine-tuned regimes. Future work includes extending the SAE analysis to intermediate LLM layers (logit lens), correlating SAE features with tampered region spatial properties, combining the frozen-feature robustness with targeted vulnerability-aware adversarial training, and using cross-backbone ensembles guided by the complementary representations identified by the CKA analysis in Part I.

Implementation Notes

Part I experiments were conducted in Google Colab; feature extraction was performed once on GPU (T4/A100) and cached as `.npy` files, with all interpretability analyses running on CPU from cached data. Part II trained all five SAE variants for 50 epochs with Adam (lr 3×10^{-4}), cosine annealing, batch size 256, decoder-column unit-norm constraint, and L1 coefficient $\lambda = 5 \times 10^{-4}$ where applicable. All intervention, boundary, and vulnerability analyses in Part II were performed by passing modified sparse codes through the SAE decoder and saved SIDA classification head, without re-running the full multimodal model.

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